Markovian Transformers for Informative Language Modeling

Scott W. Viteri Max Lamparth Peter Chatain Clark Barrett

ICLR 2026

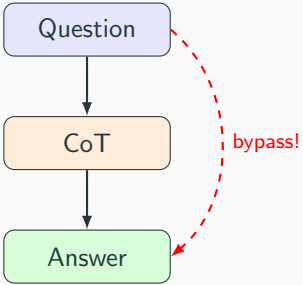
Department of Computer Science, Stanford University

The Problem: Chain-of-Thought is Not Faithful

Chain-of-Thought (CoT) prompting lets LMs "show their work" (Wei et al., 2022; Nye et al., 2022)

But CoTs can be unfaithful:

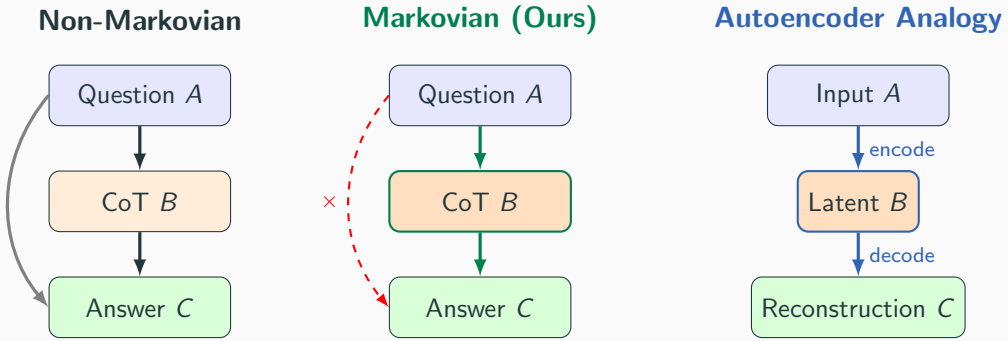
- Spurious biases remain hidden (Turpin et al., 2023)
- Altering the CoT may not change the answer (Lanham et al., 2023)
- The model can **bypass** the CoT entirely



Standard models can still "see" the question when generating the answer.

Our Approach: The Markovian Bottleneck

Key insight: remove the architectural escape hatch.



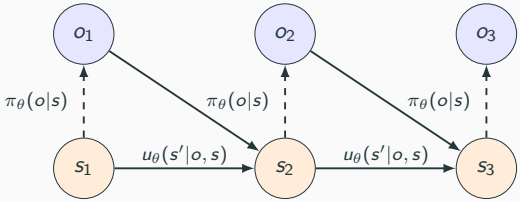
The CoT becomes a **bandwidth bottleneck**: all information from A to C must flow through B.

Formal Framework: Markovian Language Model

Definition

A **Markovian LM** is a tuple $M = (\mathcal{O}, \mathcal{S}, \pi, u, s_1)$:

- $\mathcal{O}$: observations (questions, answers)
- $\mathcal{S}$: states (CoT reasoning text)
- $\pi : \mathcal{S} \to \Delta(\mathcal{O})$: policy — predicts observations **from state alone**
- $u : \mathcal{O} \times \mathcal{S} \to \Delta(\mathcal{S})$: state update — produces CoT from question + prior state
- $s_1 \in \mathcal{S}$: initial state



Informativeness Objective & Reward

Reward: how much better is the trained CoT than the frozen baseline?

$$R_\theta = \underbrace{\ln \pi_\theta(\text{ans} \mid \text{CoT})}_{\text{trained actor on trained CoT}} - \underbrace{\ln \pi'(\text{ans} \mid \text{CoT}')}_{\text{frozen baseline on baseline CoT}}$$

Objective: maximize expected informativeness

$$J(\theta) = \mathbb{E}_{\tau \sim P, u_\theta, u'}[R_\theta(\tau)]$$

Coding-Theoretic Interpretation

- $-\log \pi_\theta(C \mid B)$: coding cost of the answer given the CoT
- $-\log u'(B \mid A)$: coding cost of the CoT under the prior
- Training searches for short states B that make both legs $A \to B \to C$ efficient

Training: GRPO-Style Policy Gradient

Key innovation: the reward R_θ depends on θ, so we get two gradient terms:

$$\nabla_\theta \mathbb{E}[R_\theta(\tau)] = \mathbb{E}\Big[\underbrace{R_\theta \cdot \nabla_\theta \ln P_\theta(\tau)}_{\text{policy gradient}} + \underbrace{\nabla_\theta R_\theta(\tau)}_{\text{actor-reward gradient}} \Big]$$

Loss function (three components):

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{PG}}}_{\text{REINFORCE}} + \underbrace{\mathcal{L}_{\text{AR}}}_{\text{direct reward}} + \underbrace{\mathcal{L}_{\text{KL}}}_{\text{regularization}}$$

- **Parallel sampling**: B diverse CoTs per question (GRPO-style)
- **Within-batch standardization**: advantages have zero mean, unit variance
- **KL penalty** ($\beta = 0.1$): discourages steganographic encodings
- **LoRA** (rank 8): stable fine-tuning with frozen base weights

Training Algorithm

Algorithm 1 Markovian Training (GRPO-Style)

Given: dataset P, actor (u_θ, π_θ), frozen baseline (u', π'), batch size B
for each training batch **do**
  Sample $(q, a) \sim P$
  Sample $\mathrm{CoT}_i \sim u_\theta(\cdot \mid q)$ for $i = 1 \dots B$ *parallel sampling*
  Sample baseline $\mathrm{CoT}' \sim u'(\cdot \mid q)$ *once per batch*
  Compute $r_i = \ln \pi_\theta(a \mid \mathrm{CoT}_i), \quad b = \ln \pi'(a \mid \mathrm{CoT}')$
  Standardize: $A_i = (r_i - b - \mu)/(\sigma + \epsilon)$
  $\ell_i = \underbrace{-\ln u_\theta(\mathrm{CoT}_i \mid q) \cdot A_i^{\mathrm{detach}}}_{\mathcal{L}_{\mathrm{PG}}} + \underbrace{(-A_i)}_{\mathcal{L}_{\mathrm{AR}}} + \underbrace{\beta_{\mathrm{KL}}\, D_{\mathrm{KL}}}_{\mathcal{L}_{\mathrm{KL}}}$
  Update θ with $\frac{1}{B}\sum_i \ell_i$
end for

Experimental Setup

Two evaluation regimes:

1. Question–Answer Tasks

- GSM8K (grade-school math)
- MMLU (multitask QA)
- SVAMP (math word problems)
- ARC-Challenge (science MCQ)
- Arithmetic (15-term addition)

CoT provides *additional computation*

2. Wikipedia Continuation

- 200 tokens context
- 50 tokens CoT
- 100 tokens target

CoT is a *bandwidth bottleneck* that must compress context

Models: Llama 3.1 8B, Mistral 7B, Qwen3 4B, Phi 3.5

Training: LoRA (rank 8, $\alpha = 16$), bfloat16, gradient clipping at 1.0

Results: QA Task Accuracy

Dataset	Base	Non-Mkv	Mkv
GSM8K	19.6%	63.3%	57.1%
ARC-Chal	36.1%	78.6%	79.9%
Arithmetic	1.0%	97.0%	98.0%
MMLU	21.4%	68.7%	55.5%
SVAMP	18.0%	43.3%	42.3%

- Large absolute gains over base model
- Within $\approx$3–4 pp of Non-Markovian
- ARC & Arithmetic: Markovian *beats* Non-Markovian
- All this while the model **cannot see the question** when answering

The Markovian constraint forces informative CoTs with minimal accuracy loss.

Perturbation Sensitivity: Is the CoT Load-Bearing?

Experiment: corrupt the CoT, measure log-probability drop for Markovian vs. Non-Markovian.

Severity	CharRep	Delete	DigRep	TruncB	TruncF	Mean
20%	+0.46	+0.46	+0.02	+0.25	−0.01	+0.24
60%	+1.04	+1.00	+0.04	+0.60	+0.28	+0.59
100%	+1.08	+1.26	+0.04	+1.26	+1.26	+0.98
Col. Mean	+0.90	+0.93	+0.03	+0.70	+0.46	+0.60

Positive = Markovian more fragile (relies more on intact CoTs). Wikipedia, 1024 examples.

Markovian models are **consistently more sensitive** to CoT corruption $\Rightarrow$ CoT is causally essential.

QA Perturbation Fragility

Dataset	CharRep	Delete	DigRep	TruncB	TruncF	Avg
ARC	+0.320	+0.424	−0.004	+0.069	+0.439	**+0.250**
SVAMP	+0.154	+0.204	+0.081	+0.076	+0.046	**+0.112**
GSM8K	+0.059	+0.069	−0.013	+0.105	+0.044	+0.053
MMLU	+0.056	+0.124	+0.004	+0.038	−0.001	+0.044
Arithmetic	−0.016	−0.003	−0.043	+0.001	−0.016	−0.015
Overall	**+0.115**	**+0.164**	**+0.005**	**+0.058**	**+0.102**	**+0.089**

Accuracy Δ (Markovian drop − Non-Markovian drop). Positive = Markovian relies more on CoT.

ARC shows the clearest Markovian fragility (+25 pp). Arithmetic is a near-tie because *both* models treat every digit as load-bearing.

Cross-Model Transfer: Wikipedia

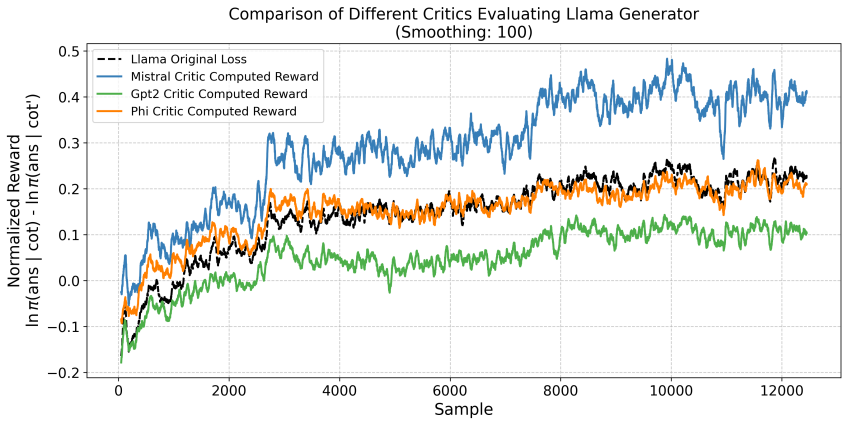


Figure 1: Llama evaluating Mistral's CoT quality during training on Wikipedia. Correlation between both models' evaluations confirms cross-architecture generalization. Averaged over 6 runs.

Qualitative Example: Arithmetic

Question: $1 + 47 + 35 + 58 + 92 + 19 + 37 + 94 + 30 + 81 + 30 + 1 + 66 + 92 + 78$

After Markovian Training (Actor):

$1 + 47 = 48$. $48 + 35 = 83$. $83 + 58 = 141$. $141 + 92 = 233$. $233 + 19 = 252$. $252 + 37 = 289$. $289 + 94 = 383$. $383 + 30 = 413$. $413 + 81 = 494$. $494 + 30 = 524$. $524 + 1 = 525$. $525 + 66 = 591$. $591 + 92 = 683$. $683 + 78 =$ **761**.

Baseline (Untrained):

To solve this problem, you need to add the numbers together step by step. You can start by adding the first two numbers, then add the result to the next number. . . (generic advice, no actual computation)

The Markovian bottleneck forces the model to actually compute in the CoT.

Qualitative Example: Wikipedia Continuation

Input:

"Apoptosis (from ...) is a form of programmed cell death that occurs in multicellular organisms and in some eukaryotic, single-celled microorganisms such as yeast. Biochemical events lead to characteristic cell changes (morphology) an"

Markovian Actor CoT:

understanding the underlying cellular processes is crucial.

Compressed text: Apoptosis is a form of programmed cell death occurring in multicellular organisms and some eukaryotic microorganisms. Biochemical events lead to cell changes an

Predicted next 50

Baseline CoT:

The text is written in a formal and technical style, which may make it difficult for some readers to understand... (metacommentary about style and tokenization)

Why Not Steganography?

Concern: could the model encode the answer steganographically in the CoT?

Three defenses:

1. **KL penalty** ($\beta = 0.1$): penalizes deviation from the pretrained CoT distribution, making unnatural encodings costly
2. **Bounded CoT length**: limits the channel capacity
3. **Inductive biases of gradient descent**: natural-language reasoning is the easier solution to discover

Empirical confirmation: cross-model transfer to GPT-2 (124M params).

- GPT-2 cannot decode sophisticated steganography
- Yet it benefits from Llama's trained CoTs
- $\Rightarrow$ CoTs use natural language, not model-specific codes

Limitations

- **Informativeness $\neq$ full faithfulness**: the CoT is sufficient for the answer, but may not mirror the model's exact internal computation
- **Human studies**: current verification uses perturbation fragility and cross-model transfer; direct human evaluation of CoTs is future work
- **Scaling**: tested up to 8B–12B parameters; larger models may behave differently
- **Post-hoc CoTs**: the model could compute the answer during question-reading passes and then generate a correct-but-post-hoc CoT (KL penalty + bounded length make this unlikely)

Summary

1. **Markovian framework**: structural constraint forcing CoT to be causally essential (answer predicted from CoT alone)
2. **GRPO-style training**: parallel sampling + actor-reward gradients + KL regularization through discrete text bottleneck
3. **Strong performance**: large gains over base (GSM8K: 19.6% $\rightarrow$ 57.1%, ARC: 36.1% $\rightarrow$ 79.9%), competitive with Non-Markovian
4. **CoTs are load-bearing**: perturbation analysis shows systematically higher sensitivity for Markovian models
5. **CoTs are interpretable**: cross-model transfer (including to GPT-2) confirms natural-language reasoning

Thank you!

Code: `github.com/scottviteri/MarkovianTraining`